Jiancong Xiao

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Academic Experience

University of Pennsylvania — Philadelphia, PA, USA
Postdoctoral Researcher — 2023 - present

- Advisors: Prof. Weijie J. Su & Prof. Qi Long

Education

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
Ph.D. in Computer and Information Engineering — 2017 - 2023

- Advisor: Prof. Zhi-Quan (Tom) Luo
- Thesis: Understanding Adversarially Robust Generalization: A Learning Theory Perspective

The Chinese University of Hong Kong — Hong Kong SAR, China
M.S. in Mathematics — 2014 - 2015

Sun Yat-sen University — Guangzhou, China
B.Econ. in Finance, Mathematics Honors Program — 2010 - 2014

- Honors program with a dual curriculum in finance and mathematics

Industry Experience

GF Futures — Guangzhou, China
Financial Engineer — 2015 - 2017

- Worked on option pricing and modeling. This experience helped clarify a strong long-term interest in theoretical research, which led to a transition to Ph.D. training.

Research Interest

I am broadly interested in **statistical learning theory** and **deep learning theory**, with a focus on developing **responsible** and **trustworthy** machine learning models. My recent research focuses on **statistical topics in large language models** (LLMs).

- **Bias and Fairness of LLM Alignment**: Analyzing algorithmic bias and fairness in preference alignment methods [J3, J2]. Studying LLM alignment through the lens of social choice theory [P1, P2]. Developing fair preference alignment algorithms.
- **LLM Training and Fine-Tuning**: Establishing theoretical foundations for fine-tuning and post-training algorithms [C2, C3, C4, W1]. Addressing calibration issues [C1] in LLMs.
- **Adversarial Robustness**: Developing theoretical frameworks for understanding adversarial examples [J4, W2], robust overfitting [C8, C6], and adversarially robust generalization [J1, C5, C7].
- **Classical Learning Theory**: Studying optimization (convergence and stability of non-convex, non-smooth problems [C8, C6, W4]) and generalization (Rademacher complexity [J1, C5], PAC-Bayes [C7, W3]).

Publications

* denotes equal contribution, † denotes alphabetical order, and ‡ denotes corresponding authors.

Journal Articles (Published or Under Revision)

[J1] **Jiancong Xiao**, Yanbo Fan, Ruoyu Sun, and Zhi-Quan Luo. "Adversarial Rademacher Complexity of Deep Neural Networks". *Journal of Machine Learning Research (**JMLR**), second revision submitted.* (2025+).

[J2] Kaizhao Liu, Qi Long, Zhekun Shi, Weijie J. Su, and **Jiancong Xiao**†. "Statistical Impossibility and Possibility of Aligning LLMs with Human Preferences: From Condorcet Paradox to Nash Equilibrium". *Annals of Statistics, major revision.* (2025+).

[J3] **Jiancong Xiao**, Ziniu Li, Xingyu Xie, Emily Getzen, Cong Fang, Qi Long, and Weijie J. Su. “On the Algorithmic Bias of Aligning Large Language Models with RLHF: Preference Collapse and Matching Regularization”. *Journal of the American Statistical Association (**JASA**)* (2024).

[J4] **Jiancong Xiao**, Liusha Yang, Yanbo Fan, Jue Wang, and Zhi-Quan Luo. “Understanding Adversarial Robustness Against On-Manifold Adversarial Examples”. *Pattern Recognition* 159 (2022), p. 111071.

Conference Papers

[C1] **Jiancong Xiao***, Bojian Hou*, Zhanliang Wang*, Ruochen Jin, Qi Long, Weijie J Su, and Li Shen. “Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach”. *Proceedings of the 42nd International Conference on Machine Learning (**ICML**)*. PMLR, 2025.

[C2] Ruochen Jin*, Bojian Hou*, **Jiancong Xiao***, Weijie J. Su, and Li Shen. “Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic”. *The Thirteenth International Conference on Learning Representations (**ICLR**)*. 2025.

[C3] Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Zhi-Quan Luo, and Ruoyu Sun. “Preserving diversity in supervised fine-tuning of large language models”. *The Thirteenth International Conference on Learning Representations (**ICLR**)*. 2025.

[C4] Mingzhi Wang, Chengdong Ma, Qizhi Chen, Linjian Meng, Yang Han, **Jiancong Xiao**, Zhaowei Zhang, Jing Huo, Weijie J Su, and Yaodong Yang. “Magnetic Mirror Descent Self-play Preference Optimization”. *The Thirteenth International Conference on Learning Representations (**ICLR**)*. 2025.

[C5] **Jiancong Xiao**, Ruoyu Sun, Qi Long, and Weijie J. Su. “Bridging the Gap: Rademacher Complexity in Robust and Standard Generalization”. *Proceedings of the Thirty Seventh Conference on Learning Theory (**COLT**)*. Vol. 247. PMLR, 2024, pp. 5074–5075.

[C6] **Jiancong Xiao***‡, Jiawei Zhang*‡, Zhi-Quan Luo, and Asuman E. Ozdaglar. “Uniformly Stable Algorithms for Adversarial Training and Beyond”. *Proceedings of the 41st International Conference on Machine Learning (**ICML**)*. Vol. 235. PMLR, 2024, pp. 54319–54340.

[C7] **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo. “PAC-Bayesian Spectrally-Normalized Bounds for Adversarially Robust Generalization”. *Advances in Neural Information Processing Systems (**NeurIPS**)*. Vol. 36. 2023, pp. 36305–36323.

[C8] **Jiancong Xiao**, Yanbo Fan, Ruoyu Sun, Jue Wang, and Zhi-Quan Luo. “Stability Analysis and Generalization Bounds of Adversarial Training”. *Advances in Neural Information Processing Systems (**NeurIPS Spotlight**)*. Vol. 35. 2022, pp. 15446–15459.

Preprints & Working Papers

[P1] **Jiancong Xiao**, Zhekun Shi, Kaizhao Liu, Qi Long, and Weijie J. Su. “Theoretical Tensions in RLHF: Reconciling Empirical Success with Inconsistencies in Social Choice Theory”. *Submitted, under review.* (2025).

[P2] Zhekun Shi, Kaizhao Liu, Qi Long, Weijie J. Su, and **Jiancong Xiao**‡. “Fundamental Limits of Game-Theoretic LLM Alignment: Smith Consistency and Preference Matching”. *Submitted, under review.* (2025).

Workshop Papers

[W1] Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo. “Entropic Distribution Matching in Supervised Fine-tuning of LLMs: Less Overfitting and Better Diversity”. *NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability (Best Paper Runner-up)*. 2024.

[W2] **Jiancong Xiao**, Zeyu Qin, Yanbo Fan, Baoyuan Wu, Jue Wang, and Zhi-Quan Luo. “Improving Adversarial Training for Multiple Perturbations through the Lens of Uniform Stability”. *ICML Workshop on New Frontiers in Adversarial Machine Learning*. 2023.

[W3] **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo. “PAC-Bayesian Adversarially Robust Generalization Bounds for Deep Neural Networks”. *ICML Workshop on New Frontiers in Adversarial Machine Learning*. 2023.

[W4] **Jiancong Xiao**, Jiawei Zhang, Zhi-Quan Luo, and Asuman E. Ozdaglar. “Smoothed-SGDmax: A Stability-Inspired Algorithm to Improve Adversarial Generalization”. *NeurIPS ML Safety Workshop*. 2022.

Grants & Funding

- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie J. Su & Prof. Qi Long (2024).
  - Assisted in proposal writing, structuring key research components.

Invited Talk

- Statistical Limits of LLM Alignment — ICSA 2025, Storrs, 2025.06
- Algorithmic Bias of RLHF — Siam MDS 2024, Atlanta, 2024.10
- Algorithmic Bias of RLHF — JSM 2024, Portland, 2024.08
- Adversarial Rademacher Complexity — COLT 2024, University of Alberta, Edmonton, 2024.06
- Stability Analysis of Adversarial Training — IOS 2024, Rice University, 2024.03
- Adversarially Robust Generalization — IOS 2024, Rice University, 2024.03
- Algorithmic Bias of RLHF — University of Pennsylvania, 2023.11
- Uniformly Stable Algorithms of Adversarial Training — MOPTA 2023, Lehigh University, 2023.08
- Stability Analysis of Adversarial Training — Sichuan University, 2023.02
- Adversarial Rademacher Complexity — Sichuan University, 2023.02
- Stability Analysis of Adversarial Training — AI Times Forum, 2022.10

Awards and Scholarships

- NeurIPS 2024 FITML Best Paper Runner-up — 2024
- NeurIPS 2023 Scholar Award — 2023
- NeurIPS 2022 Spotlight — 2022
- Poster Award of 2021 Graduate Research Conference at CUHK(SZ) — 2021
- Postgraduate Scholarship by The Chinese University of Hong Kong, Shenzhen — 2017 - 2023
- Undergraduate Scholarship by Sun Yat-sen University — 2013

Teaching

Instructor — ICSA 2025, University of Connecticut

- Statistical Inference in Large Language Models — Summer 2025

Invited Speaker — University of Pennsylvania

- STAT 9917: Statistical Topics in Large Language Models — Spring 2025

Teaching Assistant — CUHK(SZ)

- MAT 3220: Optimization II — Spring 2019 & Spring 2020
- MAT 3007: Optimization I — Fall 2019
- MAT 4001: Numerical Analysis — Fall 2018
- MAT 1004: Mathematical Analysis II — Spring 2018
- MAT 1003: Mathematical Analysis I — Fall 2017

CUHK(SZ) is a university that employs **full English medium instruction**. For the aforementioned courses, I conducted tutorial sessions in English for students, typically twice per week per course.

Professional Service

Session Organizer & Chair

- 2024 First Macau International Conference on Business Intelligence and Analytics — University of Macau, 2024.12
  - Session: Optimization and Statistics for Large Language Models
- 37th Annual Conference on Learning Theory (COLT 2024) — University of Alberta, Edmonton, 2024.06

  - Session: Adversarial and Robust Learning

- Modeling and Optimization: Theory and Applications (MOPTA 2023) — Lehigh University, PA, 2023.08
  - Session: Optimization Algorithms of Adversarial Training

Journal Reviewer

- Journal of Machine Learning Research (JMLR).
- IEEE Transactions on Information Theory (TIT).
- Journal of the American Statistical Association (JASA).
- Machine Learning.
- Quarterly Journal of Economics and Management (QJEM).

Conference Reviewer

- Machine Learning: NeurIPS 2022-2025, ICML 2023-2025, ICLR 2023-2025, AISTATS 2022.
- Computer Vision: ECCV 2024, CVPR 2023, ICCV 2021.

Mentoring

- Zhanliang Wang (on Project [C1]) — Master → Ph.D. student at University of Pennsylvania
- Ruochen Jin (on Project [C2, C1]) — Master student at ECNU → Ph.D. student at Dartmouth College
- Nikhil Vinaayak Ruia — Undergraduate student at University of Pennsylvania
- Zhekun Shi (on Project [J2]) — Undergraduate student at PKU → Ph.D. student at Princeton University
- Kaizhao Liu (on Project [J2]) — Undergraduate student at PKU → Ph.D. student at MIT
- Bohan Zhang — Undergraduate student at Peking University

Languages

- English: Fluent
- Chinese Mandarin and Cantonese: Mother tongue

Referees

Professor Weijie J. Su
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The Wharton School
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Vice President (Academic)
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